Run report: run-20261011T2333Z-bench

What ran. Five invocations of `gls_loop_bench`, each pinned to core 3 with `taskset -c 3`:
- D = −619 (`bench_run1`, start 23:33Z);
- D = −23, −71, −431, −3019 (`bench_D*`, start 23:34Z).

Each used `--pbits 127 --seed 20261012 --pairs 256 --rounds 30 --w 5`; the exact command is in each JSON. Source: the working tree that became checkpoint commit 54a16f6d4. rustc 1.97.0, release. Host: Intel Xeon @ 2.10 GHz cloud container, not isolated. Load averages before and after each run are in `../../bench_*_env.txt` (0.02–0.39 at start).

Arms. wNAF (w = 5); GLS-2; GLV-GLS-4 with an affine loop; GLV-GLS-4 with a projective loop; wNAF again (A/A control). Stage diagnostics: decomposition, α′(P) affine, α′(P) projective. Unit: ns per scalar multiplication, the per-round mean over 256 pairs. Median and min of 30 rounds; all per-round values are in the JSON.

Checks before timing (all passed, every run):
- all arms agree on every pair;
- ψ(P) = [μ]P and α′(P) = [λ]P;
- each decomposition ≡ k (mod r);
- the baseline matches double-and-add on 8 pairs.

Runs requested / completed. 5 / 5. An earlier unpinned smoke run, with only a few rounds while a subagent was compiling, read A/A 0.916. It was discarded as noisy before these runs and is not recorded.

D	wNAF	GLS-2	GLV-GLS-4 aff.	GLV-GLS-4 proj.	A/A
−23	141,473	82,216	71,168	67,744	138,830
−71	142,367	79,139	75,411	70,640	140,553
−431	140,578	79,960	78,460	71,722	138,122
−619	142,773	81,665	73,066	71,172	144,657
−3019	136,904	80,870	77,510	76,152	137,419

Medians, ns. Derived ratios of medians (GLS-2 / GLV-GLS-4, projective loop): 1.214, 1.120, 1.115, 1.147, 1.062. Baseline / A/A ratios: 1.019, 1.013, 1.018, 0.987, 0.996.

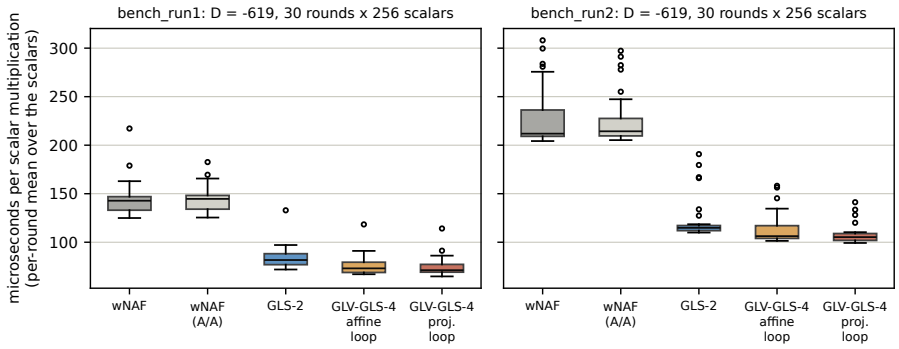


Figure 1: Per-round values, D = −619, 30 rounds × 256 scalars (run-20261011T2333Z-bench, bench_run1; a later confirmation run appears beside it if recorded).

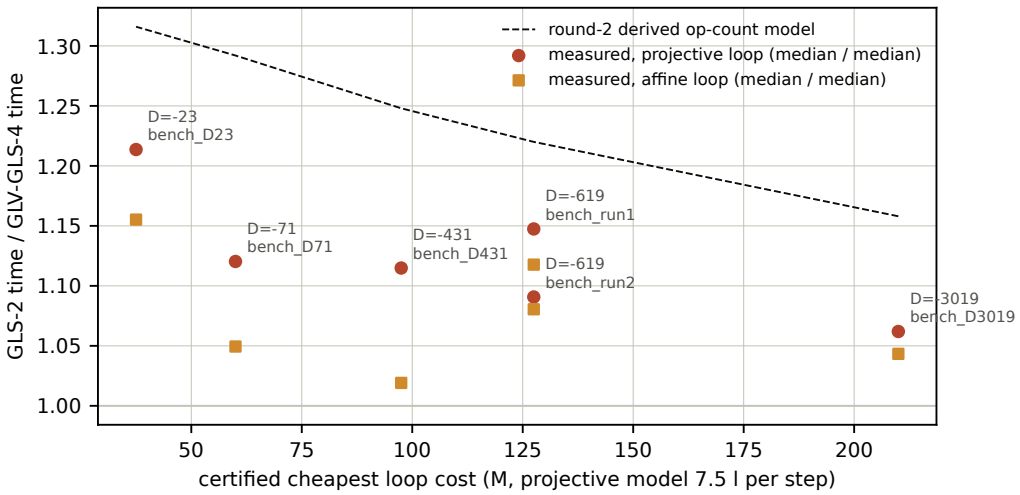


Figure 2: Ratio of recorded medians against loop model cost, with round 2′s derived model.

Observations only. Interpretation: `../../README.md` §3.